Answer Key - Engineering - Graduate Level Assessment 6 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) v(0^+) $= -4, (dv/dt)\_(0)+ = 2.25; B) v(0^+) = -2, (dv/dt)\_(0)+ = 3.25; C) v(0^+) = -3, (dv/dt)\_(0)+ = 4.25; D) v(0^+) = 0, (dv/dt)\_(0)+ = 1.5; E) v(0^+) = -3, (dv/dt)\_(0)+ = 1.25$
   Note:$Correctoption: A - v(0^+) = -4, (dv/dt)\_(0)+ = 2.25$

1. **Answer:** D
   Options: A) 0.785 J; B) 0.211 J; C) 0.125 J; D) 0.623 J; E) 0.956 J
   Note: Correct option: D - 0.623 J

2. **Answer:** D
   Options: A) a' = 2 RT_cV_c, b' = (2*V_c); B) a' = P_cV_c^2, b' = V_c / 3; C) a' = RT_c^2 / P_c, b' = V_c / P_c; D) a' = 2 RT_cV_c, b' = (V_c/ 2); E) a' = 2 RT_c, b' = V_c
   Note: Correct option: D - a' = 2 RT_cV_c, b' = (V_c/ 2)

3. **Answer:** B
   Options: A) Volt; B) Weber; C) Watt; D) Ohm; E) Joule
   Note: Correct option: B - Weber

4. **Answer:** A
   Options: A) 0.625; B) 0.683; C) 0.350; D) 1.125; E) 1.000
   Note: Correct option: A - 0.625

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '3 e^t + sin t - cos t'.

7. **Answer:** True
   Options: A) True; B) False
   Note: Matches the MMLU-Pro answer key.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'J = 1, K = 1.'.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'y [n-1] = y [n] + n^2; n $\geq$ 1 where y [0] = 0'.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 1750 Btu/hr. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** W = (AV_0) (1 / p_2 - 1 / p_1). Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key